

Pull-based consumer model

Topic

Partition

Offset

An offset in Kafka is a unique identifier for each message within a partition of a topic for Kafka consumers to track which messages they have already read.

Key

Optional

Each kafka message is a key value pair

Key can be null

Same key goes to the same partitions , kafka hash the key

If we use only one key , no scalability

Replication

Leaders and follower partitions

Idempotency for producer

Enable.idempotence = true

Each message :

Producer ID (PID)

Seq Number

(PID, partition) → sequence number

Idempotency for consumer

Check with DB (UPSERT)

Check with a cache

Implement EOS (Exactly Once Semantic)

Manual commit

Idempotent producer → protects Kafka from duplicates (write side)

EOS → protects Kafka pipeline consistency

Consumer idempotency → protects your business logic (DB, API, etc.)

Idempotency → Doing the same operation multiple times gives the same result as doing it once

Consumer group

Each consumer within a consumer group is assigned to consume from one or more partitions. This ensures that each message is processed by only one consumer in the group.

a consumer in a consumer group always reads from specific partitions assigned to it.

You shouldn't have more consumers(in a consumer group) than partitions , not efficient, extra consumers are doing nothing

How implement a queue

All consumers in the same consumer group

Consumer config → one consumer per partition

How manage ordered messages

Single partition or

Same key, so same partition, and we have offset concept ordered

lead/controller broker

Azure Event Hubs

Is Kafka

Kafka Stream

a client library for building real-time, scalable, fault-tolerant, distributed stream processing applications on top of Apache Kafka. It allows developers to process data in motion, enabling real-time analytics and event-driven microservices.